# Homework Template

Michael Joseph Ellis

November 10, 2025

## Problem 1

Double hashing with prime table size $p = 4969$. Initial hash $h(k) = k \bmod p$. Second hash $g(k) = (k+1) \bmod (p-2) = (k+1) \bmod 4967$. Probe sequence $h(k,i) = (h(k)+i \cdot g(k)) \bmod p$, $i = 0, 1, 2, \ldots$.

SSNs (in order):

$$k_1 = 132489971,\ k_2 = 509496993,\ k_3 = 546332190,\ k_4 = 034367980,$$
$$k_5 = 047900151,\ k_6 = 3229938157,\ k_7 = 212228844,\ k_8 = 325510778,$$
$$k_9 = 353354519,\ k_{10} = 053708912.$$

First hashes and steps:

| key | $h(k) = k \bmod 4969$ | $g(k) = (k+1) \bmod 4967$ |
|---|---|---|
| $k_1$ | 1524 | |
| $k_2$ | 578 | 2002 |
| $k_3$ | 578 | 1927 |
| $k_4$ | 2376 | |
| $k_5$ | 3960 | |
| $k_6$ | 3684 | |
| $k_7$ | 2854 | |
| $k_8$ | 1526 | |
| $k_9$ | 3960 | 2140 |
| $k_{10}$ | 3960 | 742 |

Insert in the given order, probing on collisions:

| key | placed at index | probes |
|---|---|---|
| $k_1$ | 1524 | 1 |
| $k_2$ | 578 | 1 |
| $k_3$ | 2505 | 2 (since 578 taken) |
| $k_4$ | 2376 | 1 |
| $k_5$ | 3960 | 1 |
| $k_6$ | 3684 | 1 |
| $k_7$ | 2854 | 1 |
| $k_8$ | 1526 | 1 |
| $k_9$ | 1131 | 2 (3960 taken) |
| $k_{10}$ | 4702 | 2 (3960 taken) |

Final locations:

$$\boxed{\begin{array}{l} k_1 \to 1524, \quad k_2 \to 578, \quad k_3 \to 2505, \quad k_4 \to 2376, \quad k_5 \to 3960, \\ k_6 \to 3684, \quad k_7 \to 2854, \quad k_8 \to 1526, \quad k_9 \to 1131, \quad k_{10} \to 4702. \end{array}}$$

## Problem 2

Power generator with $p = 7$, $d = 3$, seed $x_0 = 2$, and

$$x_{n+1} \equiv x_n^d \pmod{p}.$$

Compute:

$$x_1 \equiv 2^3 \equiv 8 \equiv 1 \pmod{7}, \qquad x_2 \equiv 1^3 \equiv 1 \pmod{7},$$

and then $x_n \equiv 1$ for all $n \geq 2$.

So the pseudorandom sequence is

$$\boxed{x_1 = 1,\ x_2 = 1,\ x_3 = 1,\ \dots}$$

(with the seed $x_0 = 2$).

## Problem 3: ISSN check digit

For digits $d_1d_2d_3d_4 - d_5d_6d_7d_8$, the check digit satisfies

$$d_8 \equiv 3d_1 + 4d_2 + 5d_3 + 6d_4 + 7d_5 + 8d_6 + 9d_7 \pmod{11},$$

and $d_8 = X$ if the residue is 10.

### (a) Compute the check digit

| Code | Weighted sum mod 11 | $d_8$ |
|---|---|---|
| $1570-868$ | $3(1) + 4(5) + 5(7) + 6(0) + 7(8) + 8(6) + 9(8) = 234 \equiv 3$ | 3 |
| $1553-734$ | $175 \equiv 10$ | $X$ |
| $1089-708$ | $218 \equiv 9$ | 9 |
| $1383-811$ | $146 \equiv 3$ | 3 |

### (b) Validity of given 8–digit strings

| Code | Required $d_8$ | Given |
|---|---|---|
| $1059-1027$ | $107 \equiv 8$ | 7 (invalid) |
| $0002-9890$ | $220 \equiv 0$ | 0 (valid) |
| $1530-8669$ | $196 \equiv 9$ | 9 (valid) |
| $1007-120X$ | $68 \equiv 2$ | $X$ (invalid) |

### (c) Single–digit errors detected

If one digit $d_j$ (including $d_8$) is changed by $\Delta \neq 0$, the congruence changes by $w_j\Delta \pmod{11}$ where $w_j \in \{3, 4, 5, 6, 7, 8, 9, 1\}$. Since none of these weights are $0 \pmod{11}$, $w_j\Delta \not\equiv 0$ and the error is always detected. Hence **all single–digit errors are detected**.

### (d) Transposition errors detected

Swapping positions $i \neq j \leq 7$ changes the checksum by $(w_i - w_j)(d_i - d_j)$. Because the weights $3, 4, 5, 6, 7, 8, 9$ are pairwise distinct mod 11, this vanishes only when $d_i = d_j$ (not an actual error). Swapping $d_7$ and $d_8$ would require $d_7 \equiv 3d_1 + \cdots + 8d_6 + 9d_8$ and $d_8 \equiv 3d_1 + \cdots + 8d_6 + 9d_7$, which imply $10(d_7 - d_8) \equiv 0 \pmod{11} \Rightarrow d_7 = d_8$. Therefore **all transpositions are detected** (except swapping identical digits, which leaves the string unchanged).

## Problem 4: Encrypt "WATCH YOUR STEP"

Use $A = 0, \ldots, Z = 25$. Ignore spaces; reinsert them in the same places.

The plaintext letters are:

W A T C H   Y O U R   S T E P

which correspond to:

$$22,\ 0,\ 19,\ 2,\ 7 \quad 24,\ 14,\ 20,\ 17 \quad 18,\ 19,\ 4,\ 15.$$

### (a)

$f(p) = (p + 14) \bmod 26$

$$\begin{array}{llll} 22 \to 10(K) & 0 \to 14(O) & 19 \to 7(H) & 2 \to 16(Q) \\ 7 \to 21(V) & 24 \to 12(M) & 14 \to 2(C) & 20 \to 8(I) \\ 17 \to 5(F) & 18 \to 6(G) & 19 \to 7(H) & 4 \to 18(S) \\ 15 \to 3(D) & & & \end{array}$$

Ciphertext:

$$\boxed{\text{KOHQV MCIF GHSD}}$$

### (b)

$f(p) = (14p + 21) \bmod 26$

$$\begin{array}{llll} 22 \to 17(R) & 0 \to 21(V) & 19 \to 1(B) & 2 \to 23(X) \\ 7 \to 15(P) & 24 \to 19(T) & 14 \to 9(J) & 20 \to 15(P) \\ 17 \to 25(Z) & 18 \to 13(N) & 19 \to 1(B) & 4 \to 25(Z) \\ 15 \to 23(X) & & & \end{array}$$

Ciphertext:

$$\boxed{\text{RVBXP TJPZ NBZX}}$$

### (c)

$f(p) = (-7p + 1) \bmod 26$

$$\begin{array}{llll} 22 \to 3(D) & 0 \to 1(B) & 19 \to 24(Y) & 2 \to 13(N) \\ 7 \to 4(E) & 24 \to 15(P) & 14 \to 7(H) & 20 \to 17(R) \\ 17 \to 12(M) & 18 \to 5(F) & 19 \to 24(Y) & 4 \to 25(Z) \\ 15 \to 0(A) & & & \end{array}$$

Ciphertext:

$$\boxed{\text{DBYNE PHRM FYZA}}$$

## Problem 5

Decrypt the messages encrypted with the shift cipher $f(p) = (p + 10) \bmod 26$. Decryption shifts back by 10.

1. `CEBBOXNOB XYG` $\to$ $\boxed{\text{SURRENDER NOW}}$.
2. `LO WI PBSOXN` $\to$ $\boxed{\text{BE MY FRIEND}}$.
3. `DSWO PYB PEX` $\to$ $\boxed{\text{TIME FOR FUN}}$.

## Problem 6

Assume a shift cipher. Try all 26 shifts; with shift $k = 4$ backward (i.e., decrypt by 4),

`ERC WYJJMGMIRXPC EHZERGIH XIGLRSPSKC MW MRHMWXMRKYMWLEFPI JVSQ QEKMG`

becomes

$\boxed{\text{ANY SUFFICIENTLY ADVANCED TECHNOLOGY IS INDISTINGUISHABLE FROM MAGIC.}}$

## Problem 7

Ciphertext: `EVIRE`. Antony will try all Caesar shifts.

With key $k = 13$ it decrypts to `RIVER` (the Tiber). With key $k = 4$ it decrypts to `ARENA` (the Coliseum).

Both are valid Caesar decodings, so the message is ambiguous without the key.

$$\boxed{\text{He can't tell; both "RIVER" and "ARENA" are possible.}}$$

## Problem 8

Compose two affine ciphers mod 26:

$$E_{a_1,b_1}(p) = a_1 p + b_1, \qquad E_{a_2,b_2}(p) = a_2 p + b_2,$$

with $a_1, a_2$ invertible mod 26. Then

$$E_{a_2,b_2}(E_{a_1,b_1}(p)) = a_2(a_1 p + b_1) + b_2 = (a_2 a_1)p + (a_2 b_1 + b_2),$$

which is again an affine map with parameters $a = a_2 a_1$ and $b = a_2 b_1 + b_2 \pmod{26}$.

$$\boxed{\text{No advantage: the composition is just another affine cipher.}}$$

## Problem 9

Using the Vigenère cipher (A=0,. . . ,Z=25). Encryption is $c_i \equiv p_i + k_i \pmod{26}$; decryption is $p_i \equiv c_i - k_i \pmod{26}$.

1. **Encrypt with key** `BLUE` the message `SNOWFALL`.

   Key shifts: B=1, L=11, U=20, E=4, repeating.
   Applying $c = p + k$ gives the ciphertext

   $$\boxed{\texttt{TYIAGLFP}}.$$

2. **Decrypt** `OIKYVWMHBX` given key `HOT`.

   Key shifts: H=7, O=14, T=19, repeating. Apply $p = c - k$ letterwise to get

   $$\boxed{\texttt{HURRHDFTIQ}}.$$

## Problem 10

Alphabet $\{a, b\}$ with $a = 0$, $b = 1$ (working mod 2). Frequencies: $P(a) = 0.1$, $P(b) = 0.9$. Ciphertext (Vigenère mod 2):

`bababaaaba`.

1. **Key length is probably 2.** Look at odd vs. even positions. Odd positions are `b b b a b` (4/5 are $b$), even positions are `a a a a a` (all $a$). With a Vigenère key of length 2 over $\mathbb{Z}_2$, each substream is either the identity or a flip. Given the heavy skew $P(b) = 0.9$, one substream being almost all $b$ and the other almost all $a$ strongly indicates a 2-period.

2. **Determine the key and decrypt.** Let the key be $(k_1, k_2) \in \{0, 1\}^2$ (letters $a, b$). Since $c = p \oplus k$, a substream with mostly $b$ must use $k = 0$ (identity), and a substream with mostly $a$ must use $k = 1$ (flip). Thus $k_1 = 0$ (odd positions), $k_2 = 1$ (even positions), i.e., key `ab`.

   Decrypt by $p = c \oplus k$:

   `b b b b b b a b b b`.

   So the plaintext is $\boxed{\texttt{bbbbbbabbb}}$ and the key is $\boxed{\texttt{ab}}$.

## Problem 11

Vigenère, key is a six-letter English word. The plaintext is the same letter repeated many times.

1. **Property revealing repeated-letter plaintext and key length 6.** If the plaintext is constant, the ciphertext is the key letters repeated periodically. Hence the ciphertext has exact period 6. A simple coincidence test (shift the text and count matches) shows huge spikes at displacements $6, 12, 18, \ldots$, revealing both that the plaintext is constant and that the key length is 6.

2. **Recovering the key once the plaintext is recognized as a single repeated letter.** Let the repeated plaintext letter be $P$ (unknown). For position $i$, $c_i \equiv P + k_i \pmod{26}$, so $k_i \equiv c_i - P \pmod{26}$. Try $P \in \{\texttt{A}, \ldots, \texttt{Z}\}$; for each $P$, the six letters $k_1, \ldots, k_6$ form a candidate key. The correct $P$ yields a six-letter English word; that is the key.

3. **If Eve just uses displacement matches, what happens? Why is the length obvious?** Let $M(d)$ be the number of matches between the ciphertext and itself shifted by $d$. Because the ciphertext is periodic with period 6, we have

$$M(d) \approx \begin{cases} N, & d \equiv 0 \pmod{6}, \\ N/26, & \text{otherwise}, \end{cases}$$

   for a long text of length $N$. Thus the plot of $M(d)$ vs. $d$ shows towering peaks at $6, 12, 18, \ldots$, making the key length unmistakable.